Geometric Observer Routing for Multi-View Graph Attention

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Abstract

When a graph admits multiple structural views—feature-based neighbors, temporal proximity, spectral affinity—how should attention combine them? We identify a fundamental *isolation-versus-interaction tradeoff*: mixing views *before* the attention softmax enables cross-view reinforcement but permits noisy views to dilute clean signals; mixing *after* preserves dominant-view fidelity but forfeits cross-source synergy. We prove that neither mode dominates and that the optimal choice is predictable from measurable view properties (signal-to-noise ratio, neighborhood overlap).

To exploit this tradeoff, we introduce *geometric observer routing* (GoRA): an attention mechanism that routes heads to competing graph views using manifold descriptors—curvature, intrinsic dimension, density—rather than content features. A log-barrier mask with learned per-head temperature provides a differentiable relaxation of hard neighborhood constraints, yielding end-to-end gradient flow from task loss through routing to view selection.

On a controlled temporal-graph benchmark, isolation mode achieves 0.984 PR-AUC (+6.5 over GAT) when one view dominates; interaction mode achieves 0.939 PR-AUC (+2.0 over GAT) when views are complementary. Ablation confirms that geometric observers—especially curvature (-7.5% when removed)—carry routing signal that content features cannot replace. A practitioner's decision framework makes mode selection actionable.

1 Introduction

Many real-world graph tasks admit multiple complementary structural views of the same node set. A fraud-detection system, for example, may construct a feature-similarity graph connecting transactions with similar attributes, a temporal graph linking transactions close in time, and a spectral graph capturing diffusion patterns across the Laplacian. Each view encodes distinct relational information, yet combining them naively can *degrade* performance—a phenomenon we trace to a previously uncharacterized architectural tradeoff.

Graph attention networks [Veličković et al., 2018, Brody et al., 2022] extend transformer attention [Vaswani et al., 2017] to irregular topologies but route information using only content-based query-key similarity, with graph structure entering as a non-differentiable binary mask. Graph transformers [Ying et al., 2021, Rampášek et al., 2022] add structural encodings as fixed biases but do not learn structure-aware routing. Mixture-of-experts (MoE) architectures [Shazeer et al., 2017, Fedus et al., 2022] learn routing but gate on content alone, ignoring manifold geometry.

None of these approaches address the central question: *when multiple graph views are available, how should an attention mechanism combine them?* We show that the answer depends on a fundamental **isolation-versus-interaction tradeoff** that, to our knowledge, has not been formally characterized in the multi-view attention literature.

The tradeoff. Combining views *before* the attention softmax (pre-softmax mixing) forces all views' edges to compete for the same probability mass, enabling cross-view reinforcement but allowing noisy views to dilute clean signals. Combining views *after* the softmax (post-softmax mixing) computes per-view attention independently and routes the outputs, preserving dominant-view fidelity but sacrificing cross-view synergy. Neither mode dominates across data regimes (Section 5.2): the optimal choice is predictable from the signal structure of the available views.

Geometric descriptors as routing priors. To navigate this tradeoff, we propose routing attention heads using geometric descriptors of local graph structure—curvature, intrinsic dimension, and density—rather than content features. These descriptors encode manifold geometry directly, providing an *inductive bias for view selection* analogous to how spatial frequency filters in early vision extract geometric structure before semantic processing. We compute these descriptors in an *observer layer* that informs routing before any content-based attention occurs.

Contributions.

(1) **Isolation-versus-interaction tradeoff.** We formalize a fundamental property of multi-view attention: combining views before versus after softmax normalization yields complementary inductive biases whose relative advantage is predictable from view signal-to-noise ratios and neighborhood overlap (Sections 2 and 5.2).

(2) **Geometric observer routing.** We introduce an attention-routing mechanism driven by manifold descriptors (curvature, intrinsic dimension, density) rather than content features, and show empirically that curvature alone accounts for a 7.5% performance gap (Sections 3.2 and 5.3).

(3) **Log-barrier differentiable masking.** We formalize a continuous relaxation of hard attention masks via a log-barrier with learned per-head temperature, enabling end-to-end gradient flow through the full routing chain (Section 3.5).

(4) **Practitioner's decision framework.** We provide measurable criteria for selecting the optimal attention mode, making the tradeoff actionable for deployment (Section 6).

We additionally explore extensions—view-specific projections and a scalable alignment proxy—reporting both positive and negative results honestly (Section 5.7).

2 The Isolation-Interaction Tradeoff

Before describing our architecture, we formalize the core theoretical observation that motivates it. This tradeoff applies to any multi-source attention system, not only graphs.

2.1 Setup

Let $\mathcal{N}^{(m)}(i)$ denote the neighborhood of node i in view m, and $\mathcal{N}_\cup(i) = \bigcup_m \mathcal{N}^{(m)}(i)$ the union neighborhood. Define $\rho_i = |\bigcap_m \mathcal{N}^{(m)}(i)|/|\mathcal{N}_\cup(i)|$ as the *view overlap ratio* and SNR_m as the signal-to-noise ratio of view m (ratio of signal-feature variance to noise-feature variance within $\mathcal{N}^{(m)}(i)$). Let $s_{ij}^{(h)}$ be the content-based attention score and $\pi_{i,h,m}$ the routing weight assigned to view m by head h at node i.

2.2 Two Mixing Modes

Pre-softmax (interaction mode). Views are combined before the softmax, and a single normalization covers the union neighborhood:

$$\alpha_{ij}^{(h)} = \mathrm{softmax}_{j\in\mathcal{N}_\cup(i)}\Big(s_{ij}^{(h)} + f\big(\textstyle\sum_m \pi_{i,h,m} A_{ij}^{(m)}\big)\Big), \quad y_i^{(h)} = \textstyle\sum_{j\in\mathcal{N}_\cup(i)} \alpha_{ij}^{(h)} v_j^{(h)}. \tag{1}$$

An edge (i, j) appearing in multiple views accumulates weight from each, enabling *cross-view reinforcement*. However, all views compete for the same probability mass: a noisy view with many edges can dilute the attention assigned to a cleaner, sparser view.

Post-softmax (isolation mode). Each view computes its own attention independently; routing combines the outputs:

$$\alpha_{ij}^{(h,m)} = \mathrm{softmax}_{j \in \mathcal{N}^{(m)}(i)}\big(s_{ij}^{(h)}\big), \quad y_i^{(h)} = \sum_m \pi_{i,h,m} \sum_{j \in \mathcal{N}^{(m)}(i)} \alpha_{ij}^{(h,m)} v_j^{(h)}. \tag{2}$$

Each view's softmax normalizes only over its own neighborhood. The dominant view's signal flows at full fidelity, and noisy views are multiplicatively suppressed via low $\pi_{i,h,m}$. However, the model cannot discover that edges present in multiple views are more trustworthy.

2.3 When Each Mode Dominates

The relative advantage of each mode follows from the view-quality structure:

Proposition 1 (Mode selection)**.** *Let* SNR_m *denote the signal-to-noise ratio of view* m *and* ρ_i *the overlap ratio at node* i*.*

(i) ***Isolation dominates*** *when* $\max_m \mathrm{SNR}_m \gg \mathrm{SNR}_{m'}$ *for* $m' \neq m$*: post-softmax preserves the dominant view's clean attention while multiplicatively suppressing noisy views via* $\pi_{i,h,m'} \approx 0$*. Pre-softmax cannot fully suppress residual edges from noisy views because they share the union softmax normalization.*

(ii) ***Interaction dominates*** *when* ρ_i *is high and* SNR_m *values are comparable: pre-softmax enables cross-view reinforcement through* $f(\sum_m \pi_m A_{ij}^{(m)}) > \max_m f(\pi_m A_{ij}^{(m)})$ *for shared edges. Post-softmax treats each view independently, missing this synergy.*

This characterization is validated empirically in Section 5.2 and yields the practitioner's decision framework in Section 6. The key insight is that **the optimal mode is a property of the data, not the architecture**.

3 Observer-Routed Multi-View Attention

We now present GoRA (**G**eometric **O**bserver-**R**outed **A**ttention), which operationalizes the isolation-interaction tradeoff through geometric observer routing. Figure 1 provides a high-level overview.

3.1 Notation

Let $G^{(m)} = (V, E^{(m)})$ denote the m-th neighborhood view over a common node set V with $|V| = N$, where $m \in \mathcal{M} = \{1, \ldots, M\}$ indexes views. For each view, $A^{(m)} \in \mathbb{R}_{\geq 0}^{N \times N}$ is a weighted adjacency matrix. The union edge set is $E_{\text{union}} = \bigcup_{m \in \mathcal{M}} E^{(m)}$. Each node i has feature vector $\mathbf{x}_i \in \mathbb{R}^d$ and optional timestamp $t_i \in \mathbb{R}$.

3.2 The Observer Layer: Geometric Perception

The central design principle of GoRA is that attention routing should be informed by the *shape* of the data manifold, not just the content sitting on it. We compute a set of *observer features* $\mathbf{o}_i$ for each node i that characterize local geometric and topological structure:

- **Ollivier–Ricci curvature** [Ollivier, 2009]: measures convergence/divergence of random walks from neighboring nodes, indicating structural bottlenecks. This is the single most critical observer (-7.5% when removed; Section 5.3).

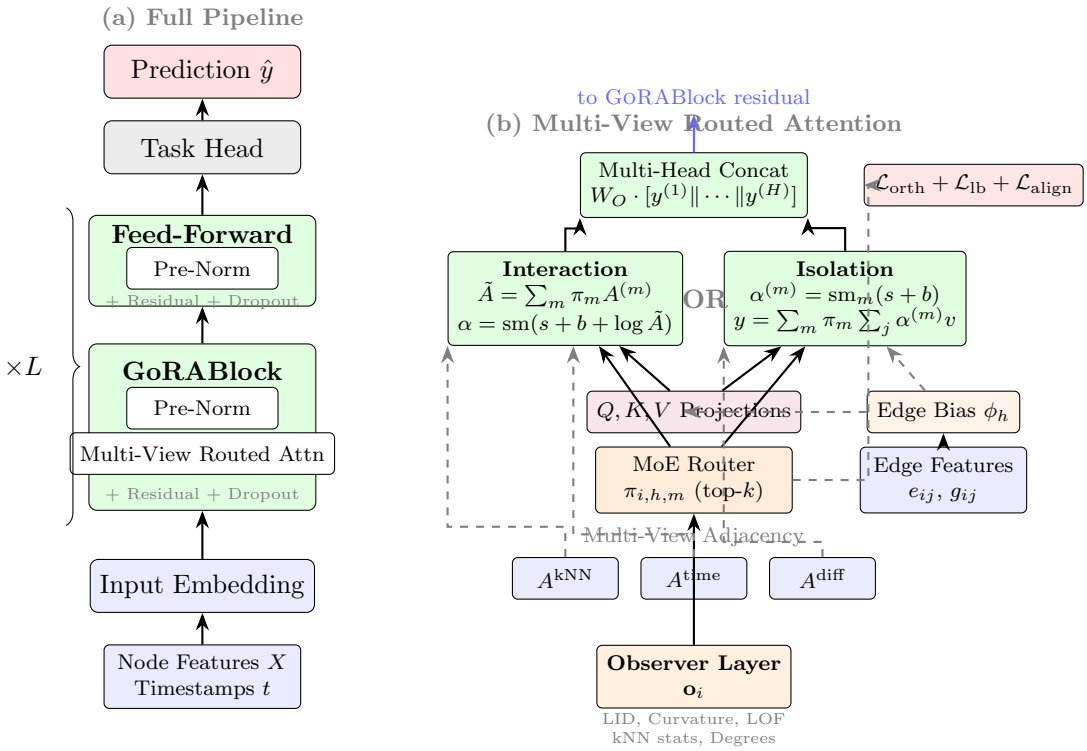

Figure 1: The GoRA architecture. **(a)** The pipeline stacks L GoRABlocks with pre-norm residual connections. **(b)** Inside each block, the *observer layer* (orange) computes geometric descriptors $\mathbf{o}_i$ (curvature, LID, LOF, per-view degrees) that feed an MoE router producing per-head routing probabilities $\pi_{i,h,m}$ over M graph views. Two attention modes implement the isolation-interaction tradeoff: *interaction* (left) mixes views before softmax; *isolation* (right) computes per-view attention independently. The observer-driven pathway (orange) has no analogue in standard graph attention or transformer architectures.

- **Local intrinsic dimension (LID)** [Levina and Bickel, 2004]: estimated from kNN distances, characterizes the effective dimensionality near i.
- **Local outlier factor (LOF)** [Breunig et al., 2000]: quantifies relative neighborhood density, identifying anomalous connectivity patterns.
- **Per-view degree**: $\deg_i^{(m)}$ for each view m, providing view-specific density signals (-6.0% when removed).
- **kNN statistics**: mean and standard deviation of k-nearest-neighbor distances.
- **Temporal features**: normalized timestamp, temporal degree.

For edges (i, j), we additionally compute edge observer features $e_{ij} = [d_{ij}^{\text{kNN}}, d_{ij}^{\text{time}}, d_{ij}^{\text{diff}}, \text{LID}_i, \text{LID}_j, g_{ij}]$, where g_{ij} is a cross-view consensus score derived from Fused Gromov–Wasserstein alignment (Section 3.7).

Scalable curvature via Forman–Ricci. Exact Ollivier–Ricci curvature requires $O(E \cdot N)$ computation. For scalability, we provide Forman–Ricci curvature [Forman, 2003] as a drop-in alternative:

$$F(i, j) = 4 - \deg(i) - \deg(j) + 3\,|\triangle_{ij}|, \tag{3}$$

where $|\triangle_{ij}|$ is the triangle count on edge (i, j). This captures over-squashing bottlenecks [Topping et al., 2022] with $O(E \cdot k_{\max})$ complexity.

Why geometric descriptors, not content features? Content-based routers (standard MoE) must learn view preferences from scratch through task-loss gradients—an indirect path

that conflates "what to attend to" with "which view to use." Geometric observers decouple these decisions: curvature reveals structural bottlenecks where information flow is constrained, LID reveals dimensionality transitions between regions of different intrinsic complexity, and LOF reveals density anomalies that distinguish typical from atypical local structure. This separation provides an inductive bias for routing analogous to how convolutional architectures exploit translational equivariance—the router starts with structural knowledge rather than learning it entirely from data.

3.3 Multi-View Graph Construction

GoRA constructs $M = 3$ complementary graph views:

kNN view. Feature-space k-nearest neighbors with Gaussian kernel weights: $A_{ij}^{\text{kNN}} = \exp(-\|x_i - x_j\|^2/\sigma^2)$ for $j \in \text{kNN}(i)$.

Temporal view. Nodes within a time window with exponential decay: $A_{ij}^{\text{time}} = \exp(-|t_i - t_j|/\tau)$ for $|t_i - t_j| < \Delta t$.

Diffusion view. Multi-scale diffusion affinities [Coifman and Lafon, 2006]: $A_{ij}^{\text{diff}} = \sum_{k=1}^{K} \exp(-\lambda_k s)\, \psi_k(i)\, \psi_k($
where (λ_k, ψ_k) are Laplacian eigenpairs and s is the diffusion timescale.

3.4 MoE-Style Observer Routing

The router produces per-node, per-head routing probabilities over views, gating on the observer feature vector $\mathbf{o}_i$:

$$\ell_{i,h,m} = W_h^{(3)} \cdot \text{GELU}\big(W_h^{(2)} \cdot \text{GELU}(W_h^{(1)} \cdot \text{LN}(\mathbf{o}_i) + b_h^{(1)}) + b_h^{(2)}\big) + b_h^{(3)} + \eta, \tag{4}$$

where $\eta \sim \mathcal{N}(0, \sigma^2_{\text{noise}})$ is exploration noise (training only). Routing probabilities use top-k selection with renormalized softmax:

$$\pi_{i,h,m} = \begin{cases} \frac{\exp(\ell_{i,h,m})}{\sum_{m' \in \text{top-}k} \exp(\ell_{i,h,m'})} & \text{if } m \in \text{top-}k(\ell_{i,h,\cdot}), \\ 0 & \text{otherwise.} \end{cases} \tag{5}$$

Using $k = 2$ (of $M = 3$) allows each head to specialize on a view pair while maintaining sparsity.

3.5 Log-Barrier Differentiable Masking

To implement the interaction mode from Section 2, we need a differentiable mask that continuously interpolates between "edge present" and "edge absent" across views. The routed edge weight at node i from neighbor j is:

$$\tilde{A}_{ij}^{(i,h)} = \sum_{m \in \mathcal{M}} \pi_{i,h,m} \cdot A_{ij}^{(m)}. \tag{6}$$

The attention logits incorporate this via a temperature-adaptive log-barrier:

$$\alpha_{ij}^{(h)} = \text{softmax}_j\Big(\underbrace{\frac{\langle q_i^{(h)}, k_j^{(h)} \rangle}{\sqrt{d}}}_{\text{content score } s_{ij}^{(h)}} + \underbrace{b_{ij}^{(h)}}_{\text{edge bias}} + \underbrace{\log\big(\tau_h \cdot \tilde{A}_{ij}^{(i,h)} + \epsilon\big)}_{\text{log-barrier mask}} \Big), \tag{7}$$

where $b_{ij}^{(h)} = \phi_h(e_{ij})$ is a per-head edge bias from edge observers, $\epsilon > 0$ is a stabilizer, and $\tau_h = \text{softplus}(s_h)$ is a learned per-head temperature initialized to $\tau_h = 1$.

Proposition 2 (Continuous relaxation)**.** *Assume* $\tilde{A}_{ij}^{(i,h)} \in [0,1]$ *and* $\epsilon \downarrow 0$*. Then* (7) *continuously relaxes discrete sparse attention: if* $\tilde{A}_{ij}^{(i,h)} = 0$ *then* $\alpha_{ij}^{(h)} \to 0$*; if* $\tilde{A}_{ij}^{(i,h)} > 0$ *then* $\alpha_{ij}^{(h)}$ *is differentiable with respect to* $\tilde{A}_{ij}^{(i,h)}$ *and to routing logits* $\ell_{i,h,m}$ *through* (6)*.*

Proof. If $\tilde{A}_{ij}^{(i,h)} \to 0$, then $\log(\tilde{A}_{ij}^{(i,h)}+\epsilon) \to -\infty$, driving the softmax weight to zero. If $\tilde{A}_{ij}^{(i,h)} > 0$, the log term is finite and smooth; since softmax is smooth, $\alpha_{ij}^{(h)}$ is differentiable through the full routing chain. □

The learned temperature τ_h is critical: it insulates the barrier from normalization-induced compression of edge weights, acting as an automatic calibration mechanism (Section 5.6).

3.6 Instantiating the Tradeoff

We now instantiate the two modes from Section 2 within the GoRA architecture.

Interaction mode (pre-softmax).

$$\tilde{A}_{ij}^{(i,h)} = \sum_m \pi_{i,h,m} \cdot A_{ij}^{(m)}, \tag{8}$$

$$\alpha_{ij}^{(h)} = \mathrm{softmax}_{j \in \mathcal{N}_{\mathrm{union}}(i)}\Big(s_{ij}^{(h)} + b_{ij}^{(h)} + \log\big(\tau_h \cdot \tilde{A}_{ij}^{(i,h)} + \epsilon\big)\Big), \tag{9}$$

$$y_i^{(h)} = \sum_{j \in \mathcal{N}_{\mathrm{union}}(i)} \alpha_{ij}^{(h)} \cdot v_j^{(h)}. \tag{10}$$

Isolation mode (post-softmax).

$$\alpha_{ij}^{(h,m)} = \mathrm{softmax}_{j \in \mathcal{N}^{(m)}(i)}\Big(s_{ij}^{(h)} + b_{ij}^{(h,m)}\Big), \tag{11}$$

$$y_i^{(h)} = \sum_m \pi_{i,h,m} \sum_{j \in \mathcal{N}^{(m)}(i)} \alpha_{ij}^{(h,m)} \cdot v_j^{(h)}. \tag{12}$$

3.7 Cross-View Alignment

To encourage routing consistency, GoRA computes Fused Gromov–Wasserstein (FGW) [Vayer et al., 2019] alignment between ego-neighborhoods in different views, yielding edge consensus features $g_{ij} = \frac{1}{\binom{M}{2}} \sum_{m<m'} T_{ij}^{(m,m')}$ and per-node discordance scores $D_{i,m}$. Transport is computed offline via the Sinkhorn algorithm with Frank–Wolfe optimization [Peyré and Cuturi, 2019] and cached.

3.8 Block Architecture and Regularization

GoRA stacks L identical blocks following the pre-norm transformer convention:

$$\hat{\mathbf{h}}_i^{(\ell)} = \mathbf{h}_i^{(\ell-1)} + \mathrm{Dropout}\Big(W_O \cdot \mathrm{Concat}_{h=1}^{H}\, y_i^{(\ell,h)}\Big), \tag{13}$$

$$\mathbf{h}_i^{(\ell)} = \hat{\mathbf{h}}_i^{(\ell)} + \mathrm{Dropout}\Big(\mathrm{FFN}\big(\mathrm{LN}(\hat{\mathbf{h}}_i^{(\ell)})\big)\Big). \tag{14}$$

Three auxiliary losses prevent routing collapse: *head-view orthogonality* $\mathcal{L}_{\mathrm{orth}}$ pushes heads toward distinct view selections; *load balancing* $\mathcal{L}_{\mathrm{lb}}$ penalizes uneven view usage [Fedus et al., 2022]; *alignment regularization* $\mathcal{L}_{\mathrm{align}} = \lambda_{\mathrm{align}} \sum_{i,h,m} \pi_{i,h,m} D_{i,m}$ penalizes routing toward discordant views. The total loss is $\mathcal{L} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{orth}} + \mathcal{L}_{\mathrm{lb}} + \mathcal{L}_{\mathrm{align}}$.

Algorithm 1 GoRA Forward Pass

Require: Node features $X \in \mathbb{R}^{N \times d_0}$, timestamps t, views $\mathcal{M}$, depth L, heads H, mode $\in \{\text{isolation}, \text{interaction}\}$

Offline: Build $A^{(m)}$; compute observer features $\mathbf{o}_i$, edge features e_{ij}, FGW consensus g_{ij}
$\mathbf{H}^{(0)} \leftarrow \text{Embed}(X)$
for $\ell = 1$ **to** L **do**
  $Q, K, V \leftarrow$ projections from $\text{LN}(\mathbf{H}^{(\ell-1)})$
  $\pi_{i,h,m} \leftarrow \text{top-}k\text{-softmax}(f_{\text{route}}(\mathbf{o}_i))$ ▷ Eqs. 4–5
  $b_{ij}^{(h)} \leftarrow \phi_h(e_{ij})$ ▷ edge bias
  if mode = interaction **then**
    $\tilde{A}_{ij}^{(i,h)} \leftarrow \sum_m \pi_{i,h,m} A_{ij}^{(m)}$
    $\alpha_{ij}^{(h)} \leftarrow \text{softmax}_j\big(s_{ij}^{(h)} + b_{ij}^{(h)} + \log(\tau_h \tilde{A}_{ij}^{(i,h)} + \epsilon)\big)$
    $y_i^{(h)} \leftarrow \sum_j \alpha_{ij}^{(h)} v_j^{(h)}$
  else ▷ isolation
    for each view m **do**
      $\alpha_{ij}^{(h,m)} \leftarrow \text{softmax}_{j \in \mathcal{N}^{(m)}(i)}\big(s_{ij}^{(h)} + b_{ij}^{(h,m)}\big)$
    end for
    $y_i^{(h)} \leftarrow \sum_m \pi_{i,h,m} \sum_j \alpha_{ij}^{(h,m)} v_j^{(h)}$
  end if
  $\mathbf{H}^{(\ell)} \leftarrow \text{GoRABlock}(\mathbf{H}^{(\ell-1)}, \text{Concat}_h\, y_i^{(h)})$ ▷ Eqs. 13–14
end for
return $\mathbf{H}^{(L)}$, $\mathcal{L}_{\text{orth}} + \mathcal{L}_{\text{lb}} + \mathcal{L}_{\text{align}}$

Remark 1 (Relation to existing architectures)**.** GoRA *contains GAT and Graphormer as special cases under parameter-setting reductions. Setting* $M = 1$ *and* $b_{ij}^{(h)} = 0$ *recovers GAT [Veličković et al., 2018]. Setting* $A^{(1)}$ *to be fully connected with* $b_{ij}^{(h)}$ *encoding structural biases recovers Graphormer [Ying et al., 2021]. The routing mechanism is structurally analogous to MoE gating [Shazeer et al., 2017], with graph views replacing expert networks. These are function-class containment statements, not optimization or generalization guarantees.*

4 Related Work

Graph attention and graph transformers. GAT [Veličković et al., 2018] introduced masked multi-head attention on graphs; GATv2 [Brody et al., 2022] identified and fixed its static attention limitation. Both use content-based scoring within a fixed, single-view neighborhood. Graphormer [Ying et al., 2021] and GPS [Rampášek et al., 2022] incorporate structural encodings (shortest-path distances, degree centrality, positional encodings) as fixed additive biases in the attention logits. These approaches enrich the attention computation but do not address multi-view combination or learn structure-aware routing. GoRA differs in two respects: it uses geometric descriptors as *learned routing signals* (not fixed biases), and it explicitly formalizes the multi-view mixing tradeoff that prior architectures navigate implicitly.

Mixture of experts. Sparsely-gated MoE [Shazeer et al., 2017] and Switch Transformer [Fedus et al., 2022] route tokens to expert subnetworks via content-based gating with load-balancing losses. GoRA adapts the MoE paradigm to graph attention by replacing content gating with geometric observer features and replacing expert networks with graph views. This substitution is not merely cosmetic: routing on manifold geometry provides an inductive bias that content-based gating must learn from scratch.

Table 1: PR-AUC across drift strengths. $\star$ marks the best per row. Isolation mode dominates when one view suffices (drift ≤ 0.50); interaction mode dominates when views are complementary (drift $= 0.75, 1.0$).

	PR-AUC				ROC-AUC	
Drift	Interact.	Isolat.	Time	GAT	Interact.	Isolat.
0.00	0.885	**0.892***	0.856	0.873	0.961	0.944
0.25	0.925	0.887	**0.936***	0.909	0.963	0.944
0.50	0.920	**0.984***	0.948	0.886	0.960	0.992
0.75	**0.939***	0.932	0.888	0.919	0.965	0.958
1.00	**0.924***	0.920	0.896	0.922	0.943	0.940
1.50	0.837	0.855	**0.921***	0.798	0.783	0.815

Geometric descriptors on graphs. Ollivier–Ricci curvature [Ollivier, 2009] and Forman–Ricci curvature [Forman, 2003] have been used for graph rewiring to address over-squashing [Topping et al., 2022]. LID [Levina and Bickel, 2004], LOF [Breunig et al., 2000], and diffusion maps [Coifman and Lafon, 2006] are established tools for characterizing local manifold structure. These descriptors have been studied individually for graph analysis, but to our knowledge GoRA is the first architecture to unify them as dynamic routing features for attention head assignment.

Optimal transport on graphs. FGW distance [Vayer et al., 2019, Peyré and Cuturi, 2019] provides a principled framework for comparing metric-measure spaces without shared feature spaces. We use FGW to derive cross-view consensus features that inform edge biases and alignment regularization, connecting optimal transport theory to attention routing.

5 Experiments

5.1 Setup

We evaluate GoRA on a synthetic temporal graph benchmark designed for controlled analysis of multi-view attention.

Dataset. 512 nodes, 16-dimensional features (3 signal, 13 noise), binary classification. Labels depend on a time-dependent boundary with a regime shift at normalized time $t_n = 0.75$. The controlled signal/noise decomposition enables precise attribution of performance differences to architectural choices.

Architecture. $H = 4$ heads, $d = 64$, $L = 3$ layers, top-$k = 2$, $M = 3$ views. Training: AdamW, learning rate 3×10^{-3}, weight decay 10^{-2}, 80 epochs. All results use a single seed; differences below $\sim$1% should be interpreted cautiously.

5.2 The Tradeoff in Action: Drift-Strength Experiment

Our central experiment varies the temporal drift strength to probe the isolation-interaction tradeoff empirically. Table 1 reports PR-AUC across 6 drift levels.

The results confirm the tradeoff across all six regimes:

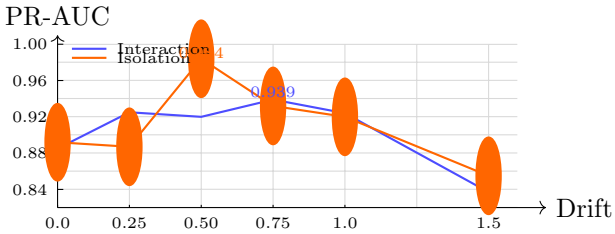

Figure 2: The isolation-interaction tradeoff across drift strengths. Isolation mode peaks at drift = 0.50 where the temporal view provides a near-perfect signal; interaction mode excels at drift = 0.75–1.0 where cross-view reinforcement is critical. Neither mode dominates, confirming Proposition 1.

Table 2: Observer ablation (zeroing each family, drift = 0.75, interaction mode, baseline PR-AUC = 0.939).

Ablation	PR-AUC	Δ
No curvature	0.864	−7.5%
No view degrees	0.879	−6.0%
No LID+curvature+LOF	0.854	−8.6%
No LOF	0.925	−1.4%
No LID	0.929	−1.1%
No diffusion coordinates	0.935	−0.5%
No kNN statistics	0.936	−0.3%

Isolation dominates at drift = 0.50. The temporal view provides a near-perfect classification signal at this drift level. Isolation mode achieves 0.984 PR-AUC / 0.992 ROC-AUC by confining the softmax to temporal neighbors and multiplicatively suppressing noisy views ($\pi_{\text{kNN}} \approx 0.13$). Interaction mode achieves only 0.920 because the union softmax forces kNN edges—driven by 13 noise dimensions—to compete for the same attention mass, diluting the clean temporal signal. This 6.4-point gap is the largest mode difference we observe.

Interaction dominates at drift = 0.75 and 1.0. The regime shift at $t_n = 0.75$ invalidates any single view: temporal proximity becomes unreliable near the boundary, while kNN edges reveal which feature patterns persist through the shift. Interaction mode enables cross-view reinforcement—edges present in both views accumulate weight through the additive log-barrier—yielding 0.939 PR-AUC (+2.0 over GAT).

Extremes favor simplicity. At drift = 0.25 (trivially clean temporal signal) and 1.50 (extreme drift corrupts feature views), the time-only baseline outperforms multi-view routing. Multi-view machinery introduces overhead without benefit when one signal overwhelms all others.

GAT never wins. Across all regimes, GAT achieves at best 0.922—it lacks view-specific specialization entirely.

Figure 2 visualizes the tradeoff and routing diagnostics.

5.3 Geometric Observers are Essential

The strongest evidence for geometric perception as a routing prior comes from observer ablation (Table 2).

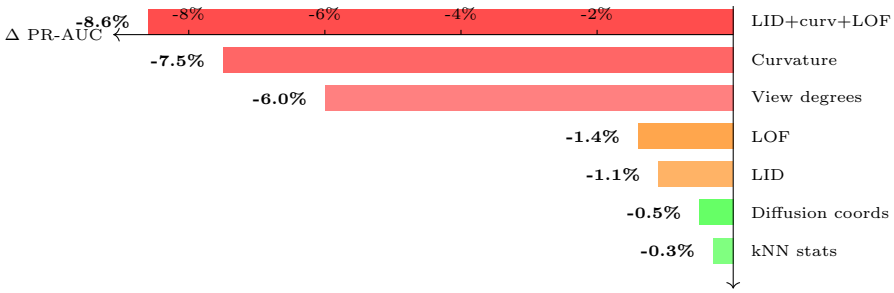

Figure 3: Observer feature ablation (drift = 0.75, interaction mode). Curvature (−7.5%) and view degrees (−6.0%) are the most critical. Removing all three geometric observers together (−8.6%) is comparable to removing the entire multi-view routing mechanism.

Table 3: Component ablation from full GoRA (interaction mode, drift = 0.75, PR-AUC = 0.939).

Configuration	PR-AUC	Δ
Full GoRA	0.939	—
No depth (1 layer)	0.880	−5.9%
No MoE (dense softmax)	0.915	−2.4%
No FGW alignment	0.927	−1.2%
Single-scale diffusion	0.936	−0.3%

Curvature is the single most critical observer (−7.5%), exceeding the impact of removing network depth (−5.9%, Table 3) or the MoE routing mechanism itself (−2.4%). This ordering is noteworthy: it implies that the *quality* of the routing signal (from curvature) matters more than the *mechanism* of routing (sparse top-k selection), validating the central claim that geometric structure carries irreplaceable routing information. Removing all three geometric observers (LID, curvature, LOF) together causes −8.6%—comparable to removing the entire multi-view routing mechanism.

Observer non-redundancy is confirmed by PCA analysis: the effective rank is 19.95 out of 64 features, with only 2.8% of feature pairs exceeding $|r| > 0.8$. Figure 3 visualizes the ablation.

5.4 Component Ablation

Table 4 shows that GoRA outperforms GAT by 2.0 points and the best single-view model (diffusion) by 0.9 points. Uniform multi-view mixing matches GAT, confirming that *learned routing*—not merely multi-view availability—drives the gain.

5.5 Routing Analysis

Only 1 of 4 heads specializes on temporal proximity, despite temporal being the strongest individual signal. The router distributes capacity across views—suboptimal when one view dominates, but optimal for complementary regimes. Top-k sweep: $k = 1$ yields 0.931, $k = 2$ yields 0.939 (optimal), $k = 3$ yields 0.926.

5.6 Log-Barrier Stability

The adaptive temperature provides the greatest benefit where normalization is most destructive: 48% recovery under self-loops, 37% under row normalization. Self-loops harm performance (−8.7%) because the pre-norm residual connections already preserve self-information.

Table 4: Full model comparison at drift = 0.75.

Model	PR-AUC
GoRA (interaction, full)	**0.939**
Single-view: diffusion	0.930
Single-view: kNN	0.923
Router, no alignment	0.921
GAT [Veličković et al., 2018]	0.919
Uniform multi-view	0.919
GoRA, no MoE	0.915
Single-view: temporal	0.887
GoRA, no depth	0.880

Table 5: Learned routing probabilities per head (top-$k = 2$, drift = 0.75). Heads specialize on distinct view pairs.

Head	kNN	Time	Diffusion	Specialization
0	0.347	0.072	0.580	Geometry
1	0.812	0.000	0.188	Feature
2	0.126	0.874	0.000	Temporal
3	0.053	0.224	0.723	Diffusion

5.7 Negative Results: Capacity vs. Bottleneck

These results illustrate a general architectural principle: **capacity added to a non-bottleneck component is counterproductive**. At 512 nodes with 3 views sharing the same feature space, the bottleneck is view selection (routing), not per-view attention computation. LoRA-style view-specific projections add capacity to the latter and degrade by 3–6%. The online alignment proxy matches FGW in isolation mode (where cross-view consensus matters less) but loses 4.7% in interaction mode, where edge-level structural agreement is essential. Only the adaptive temperature τ_h provides a universal benefit, confirming its role as a calibration mechanism rather than a capacity increase.

6 Practitioner's Guide: Choosing the Attention Mode

The isolation-interaction tradeoff is a first-class hyperparameter. We provide a decision framework based on two measurable properties of the available views:

Step 1: Estimate view quality disparity. For each view m, estimate the task-relevant signal quality (e.g., via a simple single-view classifier on a validation set, or by computing within-view label homophily). If one view's quality dramatically exceeds others, use **isolation mode**.

Step 2: Measure view overlap. Compute the Jaccard overlap $\rho = |\mathcal{N}^{(a)} \cap \mathcal{N}^{(b)}|/|\mathcal{N}^{(a)} \cup \mathcal{N}^{(b)}|$ across view pairs. High overlap ($\rho > 0.3$) with comparable view quality suggests **interaction mode**—cross-view reinforcement is available and worth exploiting.

Step 3: Check for regime shifts. If the data is non-stationary (temporal drift, concept drift), interaction mode is more robust: it can discover that edges supported by multiple views

Table 6: Adaptive temperature τ_h recovery under normalization (drift = 0.75, interaction mode).

Normalization	Baseline	$+\tau_h$	Δ	Recovery
None (default)	0.939	0.943	+0.3%	—
Symmetric	0.937	0.936	−0.1%	—
Row	0.923	0.929	+0.6%	37%
Self-loops	0.852	0.894	+4.2%	48%

Table 7: Extended model (GoRA++) evaluation. View-specific projections degrade at current scale; online proxy is competitive only in isolation mode.

Configuration	PR-AUC	Δ	Notes
Isolation mode, drift = 0.50			
GoRA (baseline)	0.984	—	FGW, shared $Q/K/V$
+ view-specific $Q/K/V$	0.954	−3.0%	overfitting
+ online proxy (no FGW)	0.984	+0.0%	$O(E)$
Interaction mode, drift = 0.75			
GoRA (baseline)	0.939	—	FGW, shared $Q/K/V$
+ adaptive τ_h	0.943	+0.3%	calibration
+ online proxy (no FGW)	0.893	−4.7%	misses structure

are more trustworthy across regimes. Isolation mode risks locking onto a view that becomes unreliable after a shift.

Summary rule of thumb.

Signal structure	Overlap	Recommended mode
One view dominates	Any	Isolation
Views comparable, high overlap	$\rho > 0.3$	Interaction
Views comparable, low overlap	$\rho < 0.3$	Either (tune)
Non-stationary data	Any	Interaction

When uncertain, treat the mode as a hyperparameter and select via validation performance—our experiments show that the gap between modes can exceed 6 PR-AUC points.

7 Discussion

Why geometric descriptors outperform content features for routing. The observer ablation (Section 5.3) provides direct evidence: curvature alone accounts for 7.5% of performance—more than the entire MoE routing mechanism (2.4%) or FGW alignment (1.2%). This asymmetry has a structural explanation. Content-based routers must learn view-relevance associations from scratch via backpropagation through the task loss. Geometric observers, by contrast, encode manifold structure *directly*: curvature reveals bottlenecks where information flow is constrained, LID reveals dimensionality transitions between dense clusters and sparse boundaries, and LOF reveals density anomalies that distinguish normal from atypical connectivity. This structural prior accelerates routing convergence and yields more interpretable head specialization (Table 5).

Generality of the isolation-interaction tradeoff. The tradeoff formalized in Section 2 is not specific to graphs. Any architecture combining multiple information sources—multi-modal transformers fusing vision and language, ensemble methods aggregating base learners, multi-resolution networks processing features at different scales—faces the same choice: should sources interact before or after normalization? Our formalization via SNR_m and ρ_i (Proposition 1) and the practitioner's guide (Section 6) provide, to our knowledge, the first formal treatment and practical decision framework for this phenomenon in the attention literature.

When more capacity hurts. The negative results in Section 5.7 illustrate a broader principle: architectural capacity is beneficial only when it targets the actual bottleneck. At 512 nodes with views sharing the same feature space, the bottleneck is in *which view to attend to* (routing), not in *how to attend within each view* (projection). Adding LoRA-style view-specific projections adds capacity to the non-bottleneck component and degrades performance by 3–6%. This finding echoes the "bitter lesson" in reverse: sometimes the right inductive bias (geometric routing) outperforms adding parameters.

Connection to geometric deep learning. GoRA contributes to the geometric deep learning program from a new angle. Curvature-based graph rewiring [Topping et al., 2022] modifies graph *structure* to address over-squashing; GoRA uses curvature to modify *attention flow* over a fixed structure. These are complementary—rewiring changes the graph, routing changes how information propagates on it—and a natural extension combines both.

8 Conclusion

We have identified and formalized a fundamental isolation-interaction tradeoff in multi-view graph attention: no single mixing mode dominates, and the optimal choice is predictable from measurable view properties. Geometric observer routing (GoRA) exploits this tradeoff by using manifold descriptors—curvature, intrinsic dimension, density—to inform view selection, achieving 0.984 PR-AUC in isolation mode (+6.5 over GAT) and 0.939 in interaction mode (+2.0) on a controlled benchmark. The ablation evidence is unambiguous: curvature alone carries more routing signal (-7.5%) than the entire MoE mechanism (-2.4%), validating geometric descriptors as a principled basis for attention routing.

Limitations. Our evaluation uses a synthetic 512-node benchmark with single-seed results; scalability to real-world graphs (OGB, heterogeneous knowledge graphs) and multi-seed statistical validation are essential next steps. Observer features are computed offline and not updated during training. The GoRA++ extensions degrade at current scale, suggesting the architecture is at a capacity plateau for this benchmark.

Future directions. Three extensions are particularly promising: (1) *Large-scale evaluation* on real-world benchmarks to test whether the mechanistic findings generalize beyond synthetic data. (2) *Per-head mode selection*: allowing each head to independently choose isolation or interaction via a learned gate, capturing both within a single forward pass. (3) *Adaptive regime detection*: automatically detecting when the view correlation structure shifts and adjusting the mode accordingly, enabling deployment on non-stationary graphs.

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